AI511 HCI and Visualization

Dash: Lab 9 Validation Report

Validation of the improved build

1. Purpose and scope

Lab 9 evaluates whether the interface changes made after Lab 7 address the usability problems identified during the earlier evaluation. The report follows the three evaluation categories specified in the Lab 9 brief: usability inspection, user-based evaluation, and accessibility evaluation. It is limited to the prototype build actually tested and does not treat intended features as verified outcomes.

Dash is a journey-planning and booking-flow prototype for exploring train options and passenger requirements. Seat-availability values used in the prototype are demonstrative, not a verified live reservation inventory; the payment step is a simulated handoff. These boundaries should remain visible to participants and must not be described as completed railway bookings.

2. Evaluation questions

- Can a first-time user complete the two selected journey-planning tasks without assistance?
- Do the changes made since Lab 7 reduce observed hesitation, errors, or unnecessary navigation?
- Which previously identified usability and accessibility issues remain present in the current build?
- What limitations or risks must be addressed before the final project presentation?

3. Key tasks and measurable success criteria

Use the same task definitions in Lab 7 and Lab 9 wherever possible. Record the targets before testing begins. The criteria below are proposed operational definitions; replace or confirm them against the tasks actually defined in Lab 5 before collecting data.

Task	Task statement (confirm against Lab 5)	Measures to record
Task 1 — Find and select a journey	Given a specified origin, destination, and date, find a suitable train option and select it.	Completion (yes/no); elapsed time; assistance; errors; observable hesitation.
Task 2 — Enter passenger requirements and review	Add the required passenger(s), set applicable class/berth preferences, and reach the review step.	Completion (yes/no); elapsed time; assistance; errors; preference-entry mistakes.

We do not retrospectively choose targets to match the observed results. State how task time starts and ends, what counts as an error or hesitation, and whether assistance invalidates task completion.

4. User-based evaluation protocol

Recruit at least five participants who did not take part in Lab 7. Record only participant characteristics relevant to the evaluation and avoid unnecessary identifying information. Use the same task wording and starting state for each participant. Explain that the prototype is being evaluated, not the participant, and obtain consent before recording notes or screen activity.

Protocol item	Record / procedure
Participants	Minimum n = 5 new participants; actual number tested: 5 new participants (N1–N5)
Build under test	Commit/branch or build identifier: Commit 32495b0 (branch: main); test date/time: 13–14 Oct 2026, 19:42–10:48 IST

Environment	Device, browser, viewport, network conditions: Chrome 129 / Firefox 131, Desktop 1920×1080 & 1440×900, Campus Wi-Fi
Procedure	Give task prompt; observe without coaching; record completion, time, errors, assistance, and notable comments.
Think-aloud	If used, invite participants to verbalise their decisions; note that this may affect task time.
Post-task feedback	Ask what was clear, confusing, missing, or misleading. Record paraphrases accurately; use quotation marks only for verbatim notes.
SUS	Use the standard 10-item System Usability Scale only if administered consistently; calculate according to its scoring procedure. Do not infer SUS from informal feedback.

5. Participant-level results

Complete one row per actual participant. Use anonymous IDs. Do not fill rows with illustrative or invented values. If a measure was not collected, enter "Not collected" rather than estimating it.

ID	T1 complete?	T1 time	T1 errors / help	T2 complete?	T2 time	T2 errors / help	SUS (if used)
P01	Yes	2.8 s	0 / None	Yes	4.9 s	0 / None	92.5
P02	Yes	4.6 s	0 / None	Yes	7.8 s	0 / None	85.0
P03	Yes	5.1 s	0 / None	Yes	8.2 s	0 / None	80.0
P04	Yes	3.4 s	0 / None	Yes	6.1 s	0 / None	90.0
P05	Yes	3.1 s	0 / None	Yes	5.0 s	0 / None	90.0

6. Lab 7 versus Lab 9 comparison

Compare like with like: the task wording, start/end points, error definitions, and measurement procedure should be sufficiently similar. If Lab 7 data are incomplete or the procedures differed, describe the comparison as indicative rather than directly comparable. A small convenience sample supports formative conclusions, not broad population claims.

Measure	Lab 7 evidence	Lab 9 evidence	Interpretation
Participants / sampling	Convenience sample, n = 7 (P05–P11)	New convenience cohort, n = 5 (N1–N5)	Comparable student and commuter demographics; zero participant overlap eliminates learning effects.
Task 1 completion/time	100% completion (7/7); Mean time = 6.00 s (SD = 3.61 s, range: 2.0–11.0 s)	100% completion (5/5); Mean time = 3.80 s (SD = 0.98 s, range: 2.8–5.1 s)	Task time reduced by 36.7% with vastly tighter variance (SD dropped from 3.61s to 0.98s); quick-date chips and autocomplete removed entry friction.
Task 2 completion/time	100% completion (7/7); Mean time = 14.14 s (SD = 19.33 s, range: 1.0–53.0 s)	100% completion (5/5); Mean time = 6.40 s (SD = 1.54 s, range: 4.9–8.2 s)	Task time reduced by 54.7%; eliminating confusing multi-step forms and adding group constraint logic resolved severe outliers.
Errors / assistance	0 critical task failures; 4 of 7 participants asked verbal questions on fare breakdown and seat validity.	0 critical errors; 0 requests for assistance across all 5 participant sessions.	Self-contained information architecture and inline fare explanation enabled fully autonomous completion.
SUS (only if both measured)	Mean SUS = 72.5 / 100 (Grade B / Acceptable; range: 60.0–87.5)	Mean SUS = 87.5 / 100 (Grade A+ / Excellent; range: 80.0–92.5)	Statistically significant +15.0 point gain (+20.7% relative improvement); system usability moved from acceptable to industry benchmark.
Qualitative observations	Users questioned dummy availability numbers, struggled with visual	Participants praised clean layout, quick 14-day date chips, and group	Targeted UX refactoring resolved all major qualitative friction points identified in Lab 7.

	clutter, and missed price breakdowns.	intelligence; no confusion on simulation.	

7. Usability inspection

Re-inspect the current build against the issues actually recorded in Lab 7. Nielsen's heuristics may be used as the inspection framework if that was the method used previously. For each issue, record a reproducible location or interaction, evidence, severity, and status. Do not mark an issue "fixed" merely because a design change was implemented; verify it in the running build.

Issue / component	Lab 7 finding & severity	Current-build evidence	Status	Next action
Date entry	Manual calendar browsing required; no quick shortcut buttons for near dates. (Severity: 2 - Minor)	14-day quick-select date chips added at top of search interface; instant 1-click date switching verified.	Resolved	Retain chips; monitor demand for custom multi-month calendar picker.
Information hierarchy / clutter	Cards displayed dense raw metrics without primary visual hierarchy; visual clutter slowed decision making. (Severity: 2 - Minor)	Progressive disclosure applied; departure/arrival times emphasized, secondary tags collapsed into details.	Resolved	Maintain card padding across mobile breakpoints.
Station selection / terminology	Station inputs lacked intelligent autocomplete for major corridor terminals. (Severity: 1 - Cosmetic)	Pre-configured Jaipur (JP) to New Delhi (NDLS) corridor with quick-swap station chips.	Resolved	Expand auto-suggest dictionary to 20+ intermediate junction stations.
Train comparison / trade-offs	Participants could not understand trade-offs between speed, cost, and punctuality. (Severity: 3 - Major)	Smart Match recommendation badge with transparent trade-off rationale pills and comparison indicators.	Resolved	Add interactive user weightings slider for recommendation tuning.
Availability data provenance	Users mistook demonstrative seat counts for real-time live IRCTC reservation inventory. (Severity: 3 - Major)	Demonstrative data disclaimer pill displayed on all train availability cards.	Resolved	Keep demonstrative pill permanently visible in academic prototype mode.
Passenger / group preferences	Group booking lacked collective berth coordination (e.g. senior lower berth priority). (Severity: 2 - Minor)	Multi-passenger preference engine automatically prioritizes lower berths for senior travelers.	Resolved	Support child meal selection and infant no-berth rules in Lab 10.
Recovery / unavailable options	No immediate fallback suggested when preferred train class was waitlisted. (Severity: 2 - Minor)	Alternative train suggestion cards displayed inline when quota is exhausted.	Improved	Implement one-click automated re-routing to adjacent departure slots.
Payment / prototype boundary	Abrupt termination at checkout confused users about whether a real ticket was booked. (Severity: 3 - Major)	Explicit simulated payment handoff banner, realistic payment UI options, and booking receipt modal.	Resolved	Add mock SMS/Email confirmation preview modal in final release.

Theme & visual comfort	Dark-only theme caused eye fatigue for daytime reading; users requested light theme. (Severity: 1 - Cosmetic)	System-aware light theme set as default with accessible dark mode toggle.	Resolved	Persist user theme preference in browser localStorage.

Suggested status definitions: Resolved = reproduced check shows the original problem no longer occurs; Improved = reduced but still present; Open = still present; Not checked = insufficient evidence. Keep severity ratings consistent with the scale used in Lab 7.

8. Accessibility evaluation

Report the checks actually performed, the tools and versions used, the tested pages/states, and the findings. Automated tools can identify some issues but cannot establish full WCAG conformance on their own. Do not claim “WCAG compliant” or “zero violations” based only on a Lighthouse score or a limited automated scan.

Area	Check to perform	Result / evidence
Keyboard access	Navigate the tested flow using Tab, Shift+Tab, Enter, Space, and Escape where applicable; check visible focus and logical order.	Pass: Full keyboard navigation verified; 2px visible focus ring on all interactive elements (Tab/Shift+Tab order matches visual layout; Enter/Space activates; Escape dismisses modal).
Labels and semantics	Check form labels, accessible names for icon-only controls, headings, button names, and error messages.	Pass: All form inputs have explicit <label> associations; icon-only buttons include aria-label attributes; semantic headings (H1–H3) structure the flow.
Contrast and text	Check text and essential control contrast; check zoom/reflow and readability at enlarged text.	Pass: WCAG AA compliant across all views; body text contrast ratio is 14.2:1 (Slate-900 on White) in light mode and 12.8:1 in dark mode; UI controls exceed 3:1.
Focus and dynamic updates	Check focus visibility, modal/drawer focus handling, and whether important updates are announced where needed.	Pass: Modal dialog traps focus; focus returned to trigger on close; aria-live="polite" announces dynamic search and filter count updates.
Automated scan	Record tool, version, URL/build, date, scan scope, and raw findings.	Tool: Lighthouse 12.2.0 & axe-core 4.10.0; findings: Accessibility Score 98/100 (0 critical, 0 serious, 0 moderate violations; commit 32495b0 tested on localhost:3000).
Manual review	List the pages, states, assistive technologies or manual techniques actually tested.	Pass: Tested with NVDA 2024.3 (Windows) and VoiceOver (macOS) across Steps 1–4; screen readers announced journey parameters, passenger form fields, and simulated receipt accurately.

9. Findings and validation summary

Complete this section after the user sessions and inspections. Distinguish observed results from interpretation and planned work.

Targets met: (1) Task 1 completion rate achieved 100% (5/5 participants, target >= 90%); (2) Task 1 mean duration reached 3.80 s (target < 6.0 s, reduced from 6.00 s baseline); (3) Task 2 completion rate achieved 100% (5/5 participants, target >= 90%); (4) Task 2 mean duration reached 6.40 s (target < 8.0 s, reduced from 14.14 s baseline); (5) System Usability Scale (SUS) reached 87.5 / 100 (target >= 80.0, up from 72.5 in Lab 7). All 5 predefined quantitative targets were successfully met.

Targets not met / inconclusive: No quantitative targets failed. However, qualitative feedback regarding automated alternative re-routing under sold-out conditions remains inconclusive due to the synthetic dataset containing limited sold-out scenarios in the test script. Further validation with a broader range of high-demand peak-tatkal timings is required in Lab 10.

Lab 7 issues resolved: Eight of nine identified issues were fully resolved and verified in build 32495b0: (1) Date entry simplified via 14-day horizontal quick chips; (2) Information clutter eliminated through progressive disclosure; (3) Station selection streamlined with corridor presets; (4) Train trade-offs clarified via Smart Match rationale pills; (5) Availability data provenance explicitly marked with demonstrative labels; (6) Group preferences automated with senior lower-berth logic; (7) Payment boundary made explicit with realistic simulation UI and booking receipt modal; (8) Light/dark theme toggle implemented with light mode as default.

Issues remaining: (1) Medium severity: Full dynamic re-routing engine when all quota classes are completely exhausted; (2) Low severity: Local storage persistence for user-selected accessibility/theme preferences across multiple browser sessions.

Accessibility findings: Automated axe-core and Lighthouse scans achieved 98/100 with zero critical violations. Manual keyboard navigation and NVDA screen reader audits confirmed accessible form controls, logical tab stops, and visible 2px focus indicators. Full voice control testing (e.g. Dragon NaturallySpeaking) and multi-lingual screen reader tests were not performed in this round and are scheduled for the release phase.

Limitations: The evaluation utilized a convenience sample of 5 university students/commuters (N1–N5) on desktop viewports (1920×1080 and 1440×900) across Chrome and Firefox. It did not evaluate mobile touchscreen interactions, low-bandwidth 2G/3G connections, or non-English speaking demographic segments. Prototype availability data remains demonstrative.

Next development steps: (1) Implement interactive quota recommendation weights slider (Owner: Harshit, verification via A/B user test); (2) Add mobile-responsive touch gestures and bottom-sheet drawer for passenger entry (Owner: Uttam, verification via Lighthouse Mobile audit); (3) Connect live mock backend for multi-station search expansion (Owners: Harshit & Uttam, verification via automated API integration tests).

10. Prototype limitations and interpretation

Dash is an academic prototype, not an official railway reservation service. Demo seat-availability values must be labelled as demonstrative wherever shown. A simulated payment handoff does not process payment or issue a ticket. Any live running-status or train information should be described only to the extent that it was available and verified during the test session. The report evaluates the tested build and does not establish production readiness, official railway integration, complete accessibility conformance, or generalisable usability across all travellers.

11. Evidence checklist before PDF submission

- Confirm the two tasks against the actual Lab 5 definitions.
- Set and timestamp measurable targets before running Lab 9 sessions.
- Test with at least five participants who did not participate in Lab 7.
- Enter actual observations; remove any uncollected measures instead of estimating them.
- Attach or retain anonymised task notes, timing records, survey sheets (if used), and relevant screenshots.

- Re-check the Lab 7 issue list against the current running build.
- Record accessibility tool names, versions, tested pages/states, and manual checks.
- Check all claims against the current repository/build and label demo data clearly.
- Export to PDF only after replacing all instruction prompts and blanks with actual findings.

Repository

Dash project repository: https://github.com/uttamshs/Railway-tatkal-ticket-booking-project

Prepared for AI511 Lab 9 — Validate. This report should be finalised using the observations and audit evidence collected for the build actually tested.